

Assignment 4

Your Name Here

A microbe called *Cinderspore* is spreading between star systems and dimming stars as it goes. One research vessel, the *Meridian*, was sent out with a small crew to study it and find a way to stop it before it reaches Earth.

The *Meridian*'s crew is busy keeping the ship running and the mission alive while you are analyzing the data sent by Frank. We have a new transmission:

Mission Log 004 — from Frank

Analyst! Hi! Okay, so, good news and weird news. Good news: the quiet-hours thing worked, mostly, I'm choosing to feel very vindicated about that. Weird news: now that Command believes me about the *two*-group stuff, they've gone and handed me *three* groups to worry about, all at once, and nobody warned me this was allowed.

Go be smarter than the two of us.

Question 0: Data Import

0a. Call in any packages that are needed for this assignment.

0b. Import this week's datasets here.

Question 1: Reflex Times

Frank: "First one's simple, I promise. We have three watch teams: Alpha, Beta, and Gamma. I keep hearing people say Gamma's reflex times are slower. But Gamma also always says *they* got the worst sleep slot! Tell me what you see"

This question will use the `reflex` dataset.

Variable	Type	Description
<code>watch_team</code>	Categorical (3 levels)	Alpha, Beta, or Gamma
<code>reflex_time_ms</code>	Continuous	Reaction time on the standard reflex drill (milliseconds)

1a. Construct the graph necessary to assess the ANOVA assumptions.

1b. Do we meet the normality assumption? Justify your answer.

Your answer here

1c. Do we meet the variance assumption? Justify your answer.

1d. Based on your responses to 1a through 1c, which hypothesis test is appropriate to determine if response times are different between the three watch groups?

Your answer here

1e. Perform the appropriate hypothesis test to determine if response times are different between the three watch groups.

Hypotheses:

- H_0 :
- H_1 :

Test Statistic and p -value

- *statistic* =
- p =

Rejection Region

- Reject H_0 if ...

Conclusion and Interpretation

- Reject or fail to reject H_0 .
- There is or is not sufficient evidence to conclude...

1f. If appropriate, perform the corresponding posthoc test that will control the Type I error rate. If a posthoc test is not appropriate, explain why.

1g. State the specific pairwise differences that exist.

Your. Answers. Here.

Question 2: Repair Times

Frank: “Reactor, Life Support, Sensor Array – pulled every repair ticket from the last quarter. Command wants to know if one of these three systems is quietly eating more of my crew’s time than the other two, because apparently ‘Frank’s gut feeling’ isn’t something they can put in a report. Rude, but fair.”

This question will use the **repair** dataset.

Variable	Type	Description
ship_system	Categorical (3 levels)	Reactor, Life Support, or Sensor Array
repair_time_hrs	Continuous	Hours logged to close the ticket

2a. Construct the graph necessary to assess the ANOVA assumptions.

2b. Do we meet the normality assumption? Justify your answer.

Your answer here

2c. Do we meet the variance assumption? Justify your answer.

2d. Based on your responses to 2a through 2c, which hypothesis test is appropriate to determine if repair times are different between the three systems?

Your answer here

2e. Perform the appropriate hypothesis test to determine if repair times are different between the three systems

Hypotheses:

- H_0 :
- H_1 :

Test Statistic and p -value

- $statistic =$
- $p =$

Rejection Region

- Reject H_0 if ...

Conclusion and Interpretation

- Reject or fail to reject H_0 .
- There is or is not sufficient evidence to conclude...

2f. If appropriate, perform the corresponding posthoc test that will control the Type I error rate. If a posthoc test is not appropriate, explain why.

2g. State the specific pairwise differences that exist.

Your. Answers. Here.

Question 3: Stress Biomarker

Frank: "Okay, this is the last one. We have stress biomarker readings from everyone's last physical. We need to know if stress is equal between the three duty bays: bridge, greenhouse, or cargo.

This question will use the stress dataset.

Variable	Type	Description
duty_bay	Categorical (3 levels)	Bridge, Greenhouse, or Cargo Bay
stress_biomarker	Continuous	Stress biomarker reading from the last physical

3a. Construct the graph necessary to assess the ANOVA assumptions.

3b. Do we meet the normality assumption? Justify your answer.

Your answer here

3c. Do we meet the variance assumption? Justify your answer.

3d. Based on your responses to 3a through 3c, which hypothesis test is appropriate to determine if stress biomarkers are different between the three duty bays?

Your answer here

3e. Perform the appropriate hypothesis test to determine if stress biomarker are different between the three duty bays

Hypotheses:

- H_0 :
- H_1 :

Test Statistic and p -value

- *statistic* =
- p =

Rejection Region

- Reject H_0 if ...

Conclusion and Interpretation

- Reject or fail to reject H_0 .
- There is or is not sufficient evidence to conclude...

3f. If appropriate, perform the corresponding posthoc test that does not control the Type I error rate. If a posthoc test is not appropriate, explain why.

3g. State the specific pairwise differences that exist.

Your. Answers. Here.